

ABDUL MUQEET KALAIGAR

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SUMMARY

Robotics / ADAS software engineer with hands-on Software-Defined Vehicle (SDV) experience on a research-grade autonomous platform — dual-board autonomy stack (NXP S32G + TI Jacinto TDA4VM / J721E), 50 Hz CAN/DBC VCU control, mmWave-radar perception, RTK-GPS fusion, and safety-critical AEB / EPB logic. Strong in ROS (Noetic + Humble), Python, C++, multimodal sensor fusion (LiDAR / RGB-D / IMU / radar), SLAM, and Jetson AGX Orin deployment. Published in Taylor & Francis (RCAAI 2025) and IEEE VTC 2026.

EXPERIENCE

TIHAN, IIT-Hyderabad — Junior Research Fellow, Robotics / ADAS R&D

Autonomous Systems • SDV • Perception

Oct 2025 – Present | Hyderabad, IN

- Own the perception ↔ VCU CAN control loop on a Software-Defined Vehicle (SDV) research platform: 50 Hz outbound drive/brake/steer frames with rolling life counters + XOR checksums; moved pacing from `sleep()` to monotonic-clock deadlines, eliminating ~10 % rate drift.
- Resolved a mid-drive parking-brake engage failure caused by overlapping life-counter streams from concurrent processes — introduced an `O_EXCL` PID-file singleton lock with stale-PID takeover and a 200 ms post-arm settle window; verified 15 s of clean 0→6 km/h motion at 0 % brake.
- Integrated a dual-board autonomy stack: **NXP S32G** (autonomy + VCU CAN, Fixposition RTK-GPS at 10 Hz) with **TI Jacinto TDA4VM / J721E** running a mmWave-radar (AWR2944) DBSCAN classifier over eight object classes at ~10 fps, with lateral / range / height gating and static-echo rejection.
- Debounced the AEB threat pipeline (N-frame threat-thold) to eliminate spurious chassis-error e-stop latches on single-frame triggers — unblocked continuous autonomous operation.
- Diagnosed and tuned the VCU steering-rate authority via the DBC steering-angle-speed signal, raising physical execution from ~100 °/s to the 500 °/s DBC maximum and lifting real steering response from 1–3 % to 70–100 % of the commanded angle within 200 ms.
- Built a Leaflet + native-OSRM waypoint-planning web UI running on the NXP S32G — multi-waypoint routing resampled at 5 m, served by a bash + nc HTTP layer on port 8080; ported OSRM native to the aarch64 embedded target (fixed 5 build errors).
- Designed the Jetson AGX Orin autonomy build + fleet-deployment pipeline: per-board device-ID fingerprinting (board serial + ECID + MAC — `/etc/machine-id` is cloneable across fleet-flashed Jetsons), Cython-locked binary protection for five core Python modules, and a 124 MB self-extracting installer with AES-256-CBC payload encryption (PBKDF2 100k iterations).
- Deployed a ROS Noetic multimodal perception stack fusing LiDAR (Livox HAP + Velodyne VLP-16) + IMU + RTK-GPS via EKF for L2+ ADAS; containerized with Docker for reproducible builds across the fleet.
- Validated navigation and perception modules in Gazebo, NVIDIA Isaac Sim, and Ansys AVxcelerate, followed by real-world testing on the PixKit research vehicle.
- Aligned safety-aware control logic (arm/disarm, e-stop, AEB, EPB) with ISO 26262 patterns for real-time automotive operation in dynamic environments.

NEWRRO Tech LLP — Robotics Training

Embedded Systems • Robotics

Nov 2024 – Dec 2024 | Bangalore, IN

- Engineered ESP32-based robotic control systems integrated with PLCs; designed CAD in Fusion 360 for hardware / software co-development.

EDUCATION

M. Tech, Robotics & AI — Nitte Meenakshi Institute of Technology, Bengaluru

Feb 2024 – Aug 2025

CGPA: 8.21

B. E., Electronics & Communication Engineering — HMS Institute of Technology, Tumkur

Aug 2018 – Feb 2022

SELECTED PROJECTS

Decentralized Navigation & Obstacle Avoidance for Swarm Robots — Multi-robot ROS2 Nav2 stack with sensor-fusion-based obstacle avoidance and cooperative path following; validated in Gazebo and on physical robots.

Pneumonia Classification & Segmentation (U-Net) — CNN classifier + U-Net segmentation on chest X-rays; **89 % classification accuracy**. Published at RCAAI 2025.

OCR-Assisted Medication Prescription Reader — Full-stack web app combining Google Vision API with a custom-trained handwriting model to extract structured medicine data from printed and handwritten prescriptions.

PUBLICATIONS

- *Identification and Classification of Pneumonia using a U-Net Algorithm* — RCAAI 2025, Taylor & Francis
- *A Real-Time Driver Drowsiness and Attention Warning System Using Adaptive Facial-Landmark Analysis and Hierarchical Alert Architecture* — IEEE VTC 2026

SKILLS

Automotive / SDV	CAN, CAN-FD, DBC, VCU control, socket-CAN, ISO 26262-familiar, safety-critical real-time systems, AEB / EPB / E-stop logic
Perception & Fusion	LiDAR (Velodyne / Livox), mmWave radar (TI AWR2944), stereo RGB-D, IMU, RTK-GPS (Fixposition Vision-RTK2), EKF, point clouds, OpenCV, PCL
Robotics & Autonomy	ROS Noetic + ROS2 Humble, Nav2, SLAM (Cartographer / RTAB-Map / ORB-SLAM3), motion & path planning, DWA, TF2
Embedded Platforms	NXP S32G, TI Jacinto TDA4VM / J721E, NVIDIA Jetson AGX Orin, ESP32, STM32, Raspberry Pi, real-time Linux
ML / CV	PyTorch, TensorFlow, TensorRT, CNNs, U-Net, YOLO, Reinforcement Learning
Simulation	Gazebo, NVIDIA Isaac Sim, Ansys AVxcelerate
Languages	Python, C++, SQL, Bash
DevOps & Tools	Docker, Git / GitHub, Linux, CI/CD, Cython, AES-256-CBC installer packaging

CERTIFICATIONS

- ROS2 (Udemy)
- Smart India Hackathon 2024
- IEEE RAS Student Summit